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Universal Modular Strategy for constructing anticoagulant coatings on surfaces of materials of medical devices

① Novelty

the authors propose a universal modular strategy for anti-coagulant strategies. The main novelty is this new system wherein multiple substrates and coatings can be accomodated. the main 2 modules are:
1. Surface Activation
2. Anticoagulant functionalisation.
Here the 12 substrates tested are from a variety of material types and allow for the creation of blood-contact material a lot more easily than usual. Technically this is not completely new because techniques such as as polydopamine and silanisation exist but its is still very significant and functional

② Scientific Rigor and methodology

the methodology is quite comprehensive because the authors investigate the coating at many levels using a lot of material characterisation techniques for example XPS, contact angle (for hydrophobicity) and amino acid density are measured. Protein adsorption & platelet adhesion are measured. Overall the science is very rigourous. Stability was tested after 18 hours and overall it was shown that the 70% of substrates retained their amino groups during this period. However in general these anticoagulant must last much longer and in the future studies must study these long terms effects.

② But they science is not perfect. Shaking does not simulate blood flow. The large scale experiments suggest heparinisation systemically this is not an ideal situation as this can't exist in the production grade system.

③ Quality of Results.

The results, as they should, ~~the rest~~ generally agree with the effectiveness of the molecules strategy. Interesting on BSA the phosphorylcholine reduced bovine serum albumin by 75% on most polymer substrates while 10% on metals & ceramics.

Therefore even though the method is universal it doesn't produce identical performance on all the substrates. In vivo definitely looks promising but I personally want to see stronger results before coming to any conclusions.

CEDD shows that embolism is adequately protected against but how much of this we can attribute to the heparin coating is questionable. PTT has a huge limitation of measurement of 2000sec so comparison between cannot be made precisely.

④ Mechanistic Understanding

The 2 coating systems are fundamentally different. Firstly an initial PEI layer ~~core~~ provides a common chemical interface between substrate & the anti-coagulant molecules

MPC is attached
PMPC-co-G
(A) B Phosph
adsor
(B) Heparin
covalent
② Limitations

I think
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briefly

1. Influence
2. It is
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5. Long

MPC is attached through a Michael addition rxn. while PMPC-co-GMA uses epoxy groups.

(A) B Phosphorylcholine → only specific protein adsorption

(B) Heparin → interfere directly with the coagulant cascade. Both mechanisms are fairly simple

(2) Limitations & gaps.

I think through my passage ①,②,③,④ I have simultaneously discussed the limitations but I will briefly list them here.

1. Influence of underlying material is not discussed
2. It is not compared with existing technologies.
3. Using microscopy we can get a better view that XPS ~~coatings~~ characterisation
4. We don't isolate the contribution of the coating.
5. Long term durability has to be studied.

NotebookLM discussion

Question 1:: Okay I understand that they have two modules and the anticoagulant attaches to PEI. But what exactly is holding the first PEI layer onto the base surface? Is that also covalent or are we relying on electrostatic forces?

The two interfaces should be separated. The authors first activate the underlying material, using plasma or acid/alkali treatment depending on the substrate, and then deposit polyethyleneimine (PEI). They describe interactions between PEI and the activated surface. The subsequently introduced anticoagulant functionality is chemically linked through reactive groups on PEI. The covalent bond in the second step does not by itself establish a covalent bond between the PEI and the original substrate (Methods 2.2; Discussion 4.1–4.2).

Bro then isnt the PEI the weak link in the entire system? If you covalently attach a really good coating to something that can come off, what have we actually achieved?

That is a useful criticism. The chemistry of the top layer can be sound, yet the device can still lose the functional coating if the PEI–substrate interface fails. A successful attachment assay on freshly coated samples does not eliminate that possibility. To demonstrate useful durability, the study would need to show that PEI and the anticoagulant remain on the device after a realistic mechanical and fluid-flow challenge, not merely that the final molecule reacts with PEI.

Wait they do say at least 70% of the amino groups remain after 18 hours in PBS, right? I initially took that as proof that the coating is stable. Does that actually measure how strongly PEI is stuck on?

Figure 2D reports retained accessible amino groups after soaking in PBS. This is evidence that the functionalized surface does not immediately lose all detectable amines under that test. But it is not a direct measurement of interfacial adhesion, and a change in the amine signal need not equal the same percentage of complete PEI-layer detachment. It also does not reproduce flowing, protein-containing blood or repeated deformation. Hence it provides an initial stability check, not a months-long device lifetime.

Can you find even one experiment that directly measures delamination force? Or PEI density before and after blood flow? Please dont just infer adhesion from contact angle or XPS.

I cannot identify a direct peel/delamination-force test or long-duration blood-flow adhesion experiment in the material summarized here. The paper uses surface chemical evidence, amino-group retention in PBS (Fig. 2D), and wettability/stability observations (including Fig. S6). Those are related but different measurements. A controlled flow-loop test could measure retained surface nitrogen/PEI, retained active agent, and anticoagulant activity before and after exposure.

So when the paper says universal, is the universal thing really just that you can create an amino surface on lots of materials? Not that it will survive for years on every actual medical device?

That is the narrower and more defensible reading. The work demonstrates a shared modular design logic across twelve substrates, using material-dependent activation routes. Universality of the fabrication concept should not be confused with identical adhesion, lifetime, safety, or anticoagulant performance on every device. This distinction matters especially for flexible catheters and high-shear extracorporeal circulation components.

Question 2: The paper claims that this surface treatment does not significantly disturb the bulk mechanical properties. Give me a vibe: what are they actually measuring here, and does unchanged Young's modulus really mean the material is unaffected?

Young’s modulus measures stiffness in the elastic regime, whereas tensile strength measures the largest tensile stress sustained, and elongation at break measures how far a sample can stretch before failure. These are different properties. The paper evaluates modulus, tensile strength, and elongation at break for several substrates (Table S1), plus flexural measurements for alumina (Table S2). Saying the surface

treatment leaves the modulus approximately unchanged would be narrower than saying it leaves all relevant mechanical behavior unchanged.

Hold on. The PP elongation goes from about 164% to 17% after plasma treatment? That isnt a small change at all. Did I read Table S1 correctly?

Those are the values reported in the supplied supplementary-table summary: untreated PP 164.3 ± 69.3%, plasma-treated PP 16.7 ± 1.7%. A dramatic decrease in extension before failure is potentially important, even if the modulus and tensile strength are relatively similar. There is also a very large change reported for PET after acid treatment, from 216.8 ± 299.1% to 7.3 ± 0.8%, although the enormous variability in the untreated PET result makes that comparison harder to interpret without raw data.

Then how can the authors still say the mechanical properties are not affected? Is it just because modulus and tensile strength are okay? I feel like that is choosing which metrics matter.

The discussion should be read precisely rather than as a blanket claim. The summarized text emphasizes that modulus and tensile strength are not significantly changed and notes variation in elongation at break. That distinction is crucial: stiffness and maximum stress may be maintained while ductility falls sharply. If someone summarizes all of this as 'mechanically harmless,' the summary would hide an important trade-off. Table S1 is the correct place to verify which substrate and pretreatment each value belongs to.

But maybe it doesn't matter? A rigid blood-pump housing never needs to stretch by 160%. Would it actually be a problem for a catheter or a flexible blood-contacting tube?

The answer depends on how the device is used. A rigid component may experience very little tensile strain, so loss of elongation might not govern its failure. A flexible catheter, mesh, or tube can be bent, stretched, crimped, or repeatedly deformed; its fatigue behavior and crack resistance may depend heavily on ductility. A universal surface-treatment platform therefore still needs device-specific acceptance criteria. Table S1 alone cannot establish cyclic durability under actual service conditions.

And do all 12 materials get the same treatment? Or is the word universal hiding the fact that they have to optimize it separately anyway?

They do not receive one identical surface pretreatment. The study uses plasma, acid, or alkali routes according to the substrate; the materials and conditions are summarized in Table S4. What is shared is the two-stage modular strategy, not a single chemical recipe. This is a useful form of versatility, but it makes it harder to claim that the mechanical side effects will be the same everywhere.

What experiment would actually settle this? Not just repeat a tensile test once. I want something that connects the chemistry to an actual working device.

For a flexible device, I would first compare untreated, pretreated-only, PEI-coated, and fully anticoagulant-coated samples. Then run cyclic bending or extension under physiological fluid exposure, measuring mechanical fatigue, cracks, coating retention, and clotting performance at intervals. That would separate damage from the activation step from damage caused by later coating chemistry. The main study does not provide that complete device-specific fatigue assessment.

Q3 give me a vibe of how the phosphorylcholine coating works. Is it basically attracting a layer of water so blood proteins cannot stick to the surface? Also why are they measuring BSA of all things?

Phosphorylcholine groups are zwitterionic and highly hydrated. A stable hydration layer can reduce nonspecific adsorption of proteins to a foreign surface; that can help reduce later platelet adhesion and activation. Bovine serum albumin (BSA) is a convenient model protein for evaluating antifouling, but it is only one species in a complicated mixture. The paper uses BSA adsorption (Fig. 3D) and includes platelet-adhesion imaging on polycarbonate (PC; Fig. 3E), so it provides more than a single protein test—but not a comprehensive thrombotic profile for every substrate.

But albumin isn't even the main culprit in a clot, right? Why not fibrinogen, von Willebrand factor, platelets under flow, or actual blood?

Correct that BSA adsorption should not be treated as a direct measure of thrombosis. Fibrinogen and von Willebrand factor can be important in platelet attachment, particularly under flow, while whole blood introduces platelets, coagulation proteins, cells, and shear. The study reports useful model assays, but the material summarized here does not show a comparable fibrinogen/vWF adsorption panel and whole-blood thrombosis test on all twelve substrates. Evidence for protein resistance is not identical to evidence for clinical clot prevention.

I also noticed the BSA reduction isn't the same on all the materials. It is above 50% for most polymers but only around 10 to 40% for the metals and ceramics? So is the chemistry universal but the result isn't?

That is an important distinction. According to Fig. 3D as summarized in the supplied assignment material, most polymer surfaces show over 50% reduction in BSA adsorption, whereas metals and ceramics show smaller reductions of roughly 10–40%. This supports fabrication across different classes of materials but not equal antifouling performance. It could reflect differences in surface chemistry, grafting density, or coating organization, but the reported reduction alone does not isolate which mechanism dominates.

Then what does Fig 3E actually prove? If platelet adhesion is shown on PC, shouldn't they show it on the metals too before saying they have a universal anticoagulant surface?

Figure 3E provides a more biologically relevant confirmation on PC than BSA adsorption alone. It does not establish that every metal, ceramic, or polymer has identical platelet behavior. An informative next test would compare the same coating chemistry across representative substrate classes in flowing whole blood, while reporting platelet coverage, thrombin formation, and the retained coating density. The resulting effect sizes could then be compared to the BSA data to see whether the surrogate predicts performance.

Question4 :The second coating strategy is heparin. They report a surface density around 0.19 to 0.66 micrograms per square centimetre. My assumption was more heparin should mean a longer clotting time. Does that relationship actually show up?

Not necessarily. A surface-density assay measures how much heparin-like material is detected, whereas anticoagulant activity depends on which heparin sequences remain accessible and can interact with antithrombin and coagulation factors. The study measures heparin density using toluidine blue (TBO; Fig. 4C) and tests coagulation separately (Fig. 4D–F). The results do not establish a calibrated, monotonic density-to-clotting-time relation for every substrate.

How does heparin actually stop clotting? And does immobilising the molecule on a solid surface interfere with the mechanism compared to free heparin in the blood?

Heparin promotes antithrombin-mediated inhibition of coagulation enzymes, notably factor Xa and thrombin, although the details depend on heparin chain length and the available binding sequence. When heparin is tethered to a surface, its chemistry, orientation, steric accessibility, and surrounding proteins can change how many active sites actually participate. The authors' end-point attachment strategy is intended to preserve anticoagulant function, but its effectiveness must be established by functional tests rather than assuming every attached chain is equally active.

So TBO dye doesn't know whether the heparin is biologically active? It basically just tells us something has attached?

TBO quantifies a dye-binding signal associated with immobilized heparin; it is not a site-specific antithrombin-binding assay. That distinction is why the functional coagulation measurements matter. A stronger characterization would report both surface loading and a direct measure of accessible anticoagulant activity after controlled aging or blood-flow exposure.

Wait the PTT is over 300 seconds in every coated sample. Is that literally the maximum that the machine can record? If everything hits the cap how on earth are we comparing the samples?

Figure 4F reports values above an upper measurement limit of 300 s. Those are censored values: they show that clotting was delayed beyond the assay window, but they cannot distinguish a true value of 320 s from 700 s. Related tests in the supplement use different ceilings (for example, the PEEK-mesh PTT result exceeds 500 s in Fig. S14). These results demonstrate a strong anticoagulant response, but do not provide an exact ranking once the ceiling is crossed.

Could we just dilute the heparin coating until the PTT comes back into range and make a proper density-versus-activity graph? Or would that be a bad experiment?

A loading-series experiment could be valuable, provided each surface is tested with the same chemistry, exposure area, blood/plasma protocol, and assay conditions. The study could extend the measurable range by selecting appropriate assay conditions or additional endpoints rather than relying on one capped test. Importantly, the independent variable should be measured active heparin loading, not only the amount of reagent used during coating.

One more thing: how long do these bound heparin groups stay active in flowing blood? An 18-hour PBS test doesn't sound like the same problem at all.

The available short-term PBS and coagulation measurements cannot answer the lifetime question under sustained blood shear. Protein fouling, loss of coating, heparin accessibility, and mechanical stress can all change with exposure. A time-resolved flow-loop experiment measuring active-agent retention plus functional clotting endpoints at multiple time points would be the next step. The paper establishes proof of concept rather than clinical-duration coating survival.

Question 5: Okay this is probably my main criticism. They say the system works in large animals. In the sheep pump experiment they still give systemic heparin, don't they? So how can we know that the MPC coating is responsible for the reduction in clots?

The sheep protocol described in Methods 2.23 includes systemic heparin at 100 U/kg. The comparison between coated and uncoated pump surfaces remains informative if the other conditions were controlled: thrombus was observed on uncoated surfaces and not obviously on coated ones after seven days (Fig. 5). But it does not establish what the coating would do without systemic anticoagulation, nor does visual absence of obvious thrombus provide a fully quantitative measure of its separate contribution.

So the control still tells us something, but not the whole story. Why didn't they just run the sheep without systemic heparin? Is that the experiment they actually need?

Withholding standard anticoagulation in an animal blood-circulation setup can carry serious welfare and experimental risks. A safer and more controlled first approach is an ex vivo blood-flow loop comparing otherwise identical coated and uncoated surfaces under defined shear and anticoagulant conditions. For any animal experiment, both groups should receive the same clinically justified systemic treatment; then the added benefit of the local coating can be estimated rather than deliberately creating an unsafe no-anticoagulation condition.

And the pig experiment is about a cerebral embolic protection device, right? Isn't the mesh basically catching the particles physically? How much of that experiment even tests heparin?

The reported pig experiment is principally a device-level test of a cerebral embolic protection device (CEPD) against injected embolic particles, described as 350–560 µm. A mesh can prevent some particles from reaching target vessels by physical filtration or deflection. That protection should be separated from the coating's anticoagulant role, which is intended to limit thrombosis on the blood-contacting device. Showing successful embolic protection is not the same as isolating the heparin coating's incremental effect.

Did they have coated CEPD versus identical uncoated CEPD? Or was it just six pigs with the device and three without? Please check the actual groups.

The groups summarized from Methods 2.23/Fig. 6 are six pigs with CEPD and three without CEPD. They are not a matched coated-CEPD versus uncoated-CEPD comparison. Thus the in vivo result supports the utility of the overall protection device but does not cleanly identify the size of the coating-specific benefit. The exact group descriptions should be checked in the paper before quoting them.

Then can we really conclude the heparin is effective in vivo? The PTT after retrieval is nice, but it sounds like a completely different claim.

The retrieved mesh's ex vivo clotting activity is supportive evidence that anticoagulant functionality persisted after the procedure. However, post-retrieval PTT measures a different endpoint from an in vivo comparison of thrombus formation on otherwise identical coated and uncoated meshes. The available observations jointly support feasibility; they do not fully separate chemical anticoagulation from physical embolic capture. Figures S13–S15 help with the activity question, but not with the missing in vivo control.

If you were the reviewer, what is the single most useful missing control? And please don't say just use more animals.

I would require identical coated and uncoated CEPDs operated under the same particle challenge, systemic anticoagulation, deployment time, flow, and device geometry. Measure two separate outcomes: embolic particle capture and coating-related thrombosis (thrombus mass or surface coverage, platelet activation, and retained heparin activity). This would reveal whether the coating adds a useful anti-thrombotic benefit beyond the mesh's purely mechanical function.

What about the supplementary material? Does it already solve this or have I missed something obvious?

The supplementary work strengthens the feasibility argument: there are blood-exposure images, heparin density/retention observations, clotting assays, and implant photographs (including Figs. S12–S15 and S19–S20, as summarized in the supplied materials). But these do not replace the missing matched in vivo coated-versus-uncoated CEPD comparison. The strongest conclusion is that the modular approach can be implemented and shows promising function, with durability and coating-specific causal attribution still needing closer testing.

Okay so my final understanding is that the paper is more a proof of concept for a modular manufacturing strategy than proof of a clinically universal, lifetime anticoagulant coating. Is that fair, or am I being too harsh?

That is a fair qualified reading. The paper contributes a broad two-module fabrication approach across diverse substrates and demonstrates meaningful chemical, in vitro, and large-animal feasibility evidence. The unresolved questions concern how uniformly it performs, how long it survives realistic use, and which portion of device performance is causally attributable to each coating. Recognizing those limits does not erase the novelty; it identifies the next experiments needed to move from platform demonstration to application-specific validation.

Reflection

A) Three specific additions to my understanding

Firstly, I understood that the word “universal” in this paper applies mainly to the modular idea, rather than identical results on all twelve substrates. The first treatment and its effect on the device still differ across materials. Secondly, I understood the distinction between a biopassive MPC coating and a bioactive heparin coating: the first tries to prevent unwanted protein/platelet attachment, whereas the second acts on the coagulation mechanism. Finally, I understood that the readouts measure different things. BSA adsorption is not a complete thrombosis test, and the amount of heparin detected by TBO is not necessarily the same as its biological activity.

B) One corrected misconception

I initially assumed that because the anticoagulant molecules were covalently attached, the entire coating was covalently bound to the medical device. But the covalent chemistry is primarily at the functional-group-to-PEI interface; the PEI-to-substrate interface must be considered separately. I also took the reported short-term amino-group retention as strong evidence of stability. It is useful evidence, but it does not establish adhesion under long-term pulsatile blood flow.

C) One answer that needed additional verification

I had to go back to the controls in the large-animal experiments rather than simply accept the general claim that the coating worked in vivo. Methods 2.23 reports systemic heparin in the sheep work. In the pig study, the comparison is CEPD versus no CEPD, rather than coated versus uncoated CEPD. Hence the results support device feasibility but cannot fully isolate the coating’s contribution. I would check the exact group descriptions in Methods 2.23 and Figures 5–6 before drawing a stronger causal conclusion. But as you can see NotebookLM is worse than GPT-4 and therefore all questions needed verification

D) Revised design recommendation

I would keep the two-module concept, but I would not treat every substrate as equivalent. For flexible devices I would choose a surface activation method that preserves elongation at break and fatigue resistance. I would also test the PEI interface and functional coating under a defined blood-flow shear challenge, measuring how much active material remains and whether its anticoagulant function decreases. Finally, I would include matched coated and uncoated devices so the contribution of the coating is separated from the functioning of the device itself.

E) One technically meaningful unanswered question

How do repeated mechanical deformation and months of pulsatile blood exposure affect the attachment of PEI to different substrate classes, and how closely does any loss of this interface predict loss of functional anticoagulant activity?